



Smart Web Ordering System with Chatbot Assistance and Generative Cake Design

AY 2025-2026

AUTHORS:

John Carlo E. Baracena
Ashton Brian S. Garcia
Charlize Jacee Garcia
Nathaniel J. Villafior

CO-AUTHORS:

Manuel L. Calimlim, Jr, PhD

CONTRIBUTES TO:

8 DECENT WORK AND
ECONOMIC GROWTH



9 INDUSTRY, INNOVATION
AND INFRASTRUCTURE



12 RESPONSIBLE
CONSUMPTION
AND PRODUCTION



INDIRECTLY AIDS:

11 SUSTAINABLE CITIES
AND COMMUNITIES



13 CLIMATE
ACTION



Table of Contents

Introduction	3
Project Context.....	3
Statement of the Problem	5
Objectives.....	5
Significance of the Project	6
Scope and Limitations.....	7
Review of Related Literature / Systems	9
Review of Related Systems.....	12
Current System	17
Technical Background.....	17
List of Processes	20
SWOT Analysis	23
Proposed Solution.....	24
Technical Background.....	24
Feasibility.....	25
Operational Feasibility	25
Economic Feasibility.....	25
Technical Feasibility	26
Schedule Feasibility.....	26
Requirements Analysis	26
Project Vision.....	26
Prototype (Mock Flow / Wireframe).....	26
Project Lean Canvas	35
User Classes and Characteristics	36
Product Roadmap.....	40
Use Case Diagram	40
Conclusion	42
References.....	44
Appendices	46
Appendix A: Roles and Responsibilities	46
Appendix B: Minutes of the Meetings	46
Appendix D: Project SharePoint Link.....	48
Appendix E: Requirements Traceability Matrix	48

List of Figures

Figure 1: Privacy Notice.....	27
Figure 2: Log in Page	27
Figure 3: About Us.....	28
Figure 4: Our Services.....	28
Figure 5: Cake Design	29
Figure 6: Home Page.....	29

List of Tables

Table 1: Social Media Ordering Process.....	21
Table 2: GrabFood & FoodPanda Ordering	22
Table 3: Walk-In Dine-In Service.....	22
Table 4: SWOT Analysis.....	24
Table 5: Online Customers.....	37
Table 6: Shop Owner and Staff	38
Table 7: Product Backlog / User Stories	39

Introduction

The growth of digital platforms has changed how small businesses operate, especially in the food industry. Many pastry businesses today rely on social media platforms such as Facebook and Instagram to promote products, communicate with customers, and accept orders. These platforms are easy to use and accessible, making them a common choice for small and starting businesses. However, relying only on social media can create challenges in managing orders, responding to inquiries, and organizing business transactions.

Project Context

Mama & Co., formerly known as Fat Mama, began in 2020 as a home-based cloud kitchen located in Villa Lourdes, Lipa City, Batangas. Initially established as a small passion project during the pandemic, the business gradually expanded through social media exposure and word-of-mouth recommendations from family and friends. Over time, it evolved into a multi-brand hybrid operation that includes concepts such as Ben & Mama and 15th Street Kitchen. The company also formed partnerships with foodpanda, GrabFood, Tapa King, and Bo's Coffee. In 2024, Mama & Co. further expanded by opening its first physical dine-in store while maintaining a strong online presence through Facebook and Instagram.

Mama & Co. offers a variety of products including custom cakes, pastries, rice meals, and beverages under its multi-brand setup. The business primarily serves customers within Villa Lourdes, Lipa City, Batangas through both dine-in and online ordering channels. Currently, customer orders are received through social media platforms such as Facebook and Instagram

direct messages, as well as food delivery platforms like Foodpanda and GrabFood. The existing ordering process follows this sequence: customer places an order via Facebook or Instagram, staff manually confirms item availability, staff records the order into the inventory application system, the order is either modified, accepted, or cancelled, a receipt is generated, order preparation begins, and the completed order is either picked up by a rider or claimed through pickup before the transaction is recorded in the POS system.

Despite the business's growth and increasing customer demand, the current manual ordering process presents significant operational and scalability challenges. Customer inquiries, order confirmations, inventory tracking, and transaction management are primarily handled manually through social media messaging platforms. Customers commonly inquire about products, pricing, and custom cake designs through direct messages, while staff manually respond to and record orders. Although this approach is accessible and convenient for customers, it becomes increasingly difficult to manage as order volume increases. The business currently handles an average of 40–50 orders daily, with peak periods reaching between 200 and 350 orders. As a result, messages may be overlooked, response times may be delayed, and orders may become disorganized. The current average social media response time ranges from 30 to 60 minutes. This existing system without a centralized platform capable of managing customer information, transactions, inventory, and orders in a unified system. In addition, customers are unable to clearly visualize custom cake designs before confirming their purchases, which may lead to misunderstandings, inaccurate expectations, and dissatisfaction. These operational inefficiencies affect both the business workflow and the overall customer experience, highlighting the need for a more organized and standardized digital solution capable of supporting the company's continuous growth.

According to Williams (2025), social media and online platforms have transformed how small businesses operate by enabling them to market products, interact with customers, and expand their reach. However, the study also emphasizes that adopting integrated digital tools is essential for businesses seeking to scale operations effectively. In line with this, the proposed project aims to develop a smart web-based ordering system with chatbot assistance and generative cake design features for Mama & Co. It seeks to automate customer inquiries, streamline order management, improve inventory tracking, and allow customers to visualize custom cake designs

before purchasing confirmation. By integrating these functionalities into a centralized web-based platform, the project aims to enhance operational efficiency, improve order accuracy, reduce response times, and provide a more convenient and satisfying customer experience.

Statement of the Problem

The client currently manages the cake business mainly through social media which poses few challenges:

1. **Difficulty in Order Tracking:** Scattered chat messages make it hard to record, monitor, and update orders, leading to inconsistent records and difficulty in managing the daily volume of customers.
2. **Time-Consuming Inquiry Handling:** The process of responding to customer inquiries is time-consuming due to repetitive questions that require manual responses every time.
3. **Customization mismatch:** Not being able to preview the cake design often causes confusion, frequent changes, and customer dissatisfaction with their orders.
4. **Problem in Inventory Tracking:** Without proper inventory tracking, the pastry shop's ordering system becomes prone to order errors and inconsistencies in product availability and customer service.
5. **Difficulty in Report Generation:** The manual system makes report generation time-consuming and error-prone, resulting in delays and unreliable information.

Objectives

The primary goal is to develop an integrated website featuring an **AI Chatbot** and **Generative AI cake design** tool to streamline order management and enhance customer interaction.

1. Improve the recording, monitoring, and updating of customer orders by organizing order information into a consistent and structured process, reducing record inconsistencies, and improving the management of daily customer volume.
2. Reduce the time spent responding to customer inquiries by standardizing responses to common questions, aiming to decrease overall response time by at least 50%
3. Reduce order confusion and design-related changes by clearly defining and confirming customer design expectations, aiming to decrease customization errors and customer dissatisfaction.
4. Improve the accuracy and consistency of stock monitoring and recording practices to reduce order errors and ensure reliable product availability, aiming to decrease inventory-related issues.
5. Improve the accuracy and efficiency of report preparation by standardizing data recording and organization practices, aiming to reduce report preparation time and errors.

Significance of the Project

The findings of this study would benefit from the following:

1. **The Owner of the Business**

The proposed ordering system with AI chatbot and generative cake design will significantly improve the business efficiency by automating order management and customer communication, reducing any manual workload, and provides a more standard platform for operations.

2. **Staff**

The system will help the staff by making their work easier and more organized. Instead of handling orders through Facebook and Instagram, it reduces confusion and errors. The automated system will also lessen their workload in answering repetitive inquiries, allowing them to focus more on preparing orders and improving service. This will lead to faster transactions and a more efficient workflow.

3. **Customers**

Customers will experience a convenient, interactive, and responsive ordering experience. The generative AI features allow them to visualize their own custom cake designs and reduce the difficulty of getting a design idea, which will raise their level of satisfaction.

4. **Future IT Developers**

This study will provide practical experience in full-stack web development and the integration of AI technologies, preparing them for real-world software engineering challenges.

Scope and Limitations

This study explores the design, development, and application of an AI-enhanced web ordering system for Mama & Co., a cloud kitchen and pastry business that relies on social media platforms for order management. The platform to be created will enhance the business operational activities, customer inquiries, and the capability of visualizing the product to be delivered to customers.

The core components of this system include: (1) an AI chatbot with NLP capabilities that automatically answers common customer queries, filters requested ingredient exclusions, and routes complex inquiries to human staff; (2) a generative AI cake design feature powered by DALL-E that produces visual previews of custom orders based on customer-provided

preferences; (3) a centralized POS portal that consolidates orders from social media and third-party delivery platforms such as FoodPanda and GrabFood; (4) an AI-driven inventory management module that automatically checks ingredient stock and suggests available alternative ingredients when items are out of stock; (5) a statistics and analytics dashboard for tracking order volume, peak seasons, and retention metrics. The system interface is organized into three clearly separated sections: the POS Portal, the Chatbot Interface, and the Generative AI Design Section.

Customer segments of interest are Mama & Co.'s customers in the Lipa City area, including young adults, families, and couples looking to purchase foods and pastries online.

Despite aiming to improve the operational process of the business, several limitations will be applied. This system was developed based on current operational needs; future changes in business operations will require additional modifications. The generative AI cake design feature is limited to prompts within Mama & Co.'s actual design capabilities—designs that are too complex, require unavailable ingredients, or involve techniques beyond the pastry shop's capacity will be flagged for staff review. Since the system operates entirely online separating to onsite shop, a constant internet connection is required; a full offline version is outside the scope of this study. A full third-party data privacy audit is not included, though the system will comply with the Data Privacy Act of 2012 (RA 10173) through password hashing, secure database hosting, a data privacy consent checkbox during checkout, and a cookie consent mechanism on the website.

Review of Related Literature / Systems

This chapter presents a review of related literature and studies, both foreign and local, that are relevant to the development of an Smart web-based ordering system for Mama & Co. The literature and studies gathered in this chapter serve as the theoretical and conceptual foundation of the present research. The reviews work cover topics including AI-powered chatbots, generative artificial intelligence, web-based ordering systems, and the digital transformations of small food businesses. These references were carefully selected to provide insight into existing technologies, methodologies, and findings that are directly related to the proposed system. By examining prior research and developments in these field, this chapter aims to establish a relevance and necessity of the present study and to identify the gaps that the proposed system seeks to address.

Foreign Literature

AI-Powered Chatbot in the Restaurant Industry

Gupta, Dheekonda, and Masum (2024) introduced "Genie," an AI-powered chatbot designed to enhance restaurant operations including order processing, reservations, and FAQ management. The system leverages advanced Natural Language Processing (NLP) techniques by converting input queries into word embeddings and applying a tag classification system to accurately interpret customer intent and generate responses. Their study found that the combination of Word2Vec and Artificial Neural Networks (ANN) proved to be the most effective approach, achieving an accuracy rate of 88.9%, highlighting the chatbot's capability to streamline restaurant operations and minimize customer wait times. This study is highly relevant to the present research as it demonstrates the practical application of AI chatbots in automating customer communications within the food industry, which directly addresses the manual inquiry management challenges faced by Mama & Co.

AI-Powered Personalization in Food Recommendations

Yaiprasert and Hidayanto (2024) proposed and evaluated a novel approach utilizing ensemble machine learning techniques for personalized meal services, addressing a critical gap in understanding AI-powered decision-making within the food delivery and restaurant industry. Their framework involves tagging food data — including food type, ingredients, nutritional content, and

cooking methods — alongside customer information such as dietary preferences and budget, enabling the system to deliver highly personalized food recommendations. This literature supports the importance of AI-driven personalization in the food industry, which relates to the present study's goal of improving customer ordering experience through intelligent and responsive system features.

AI's Impact on the Restaurant Industry

Jenkins (2024) highlighted that artificial intelligence is increasingly being adopted across the restaurant industry to improve efficiency, reduce operational costs, and enhance the overall customer experience. AI applications in the food sector range from automated ordering systems and chatbots to predictive analytics and inventory management. These developments underscore the growing need for food businesses, including small and emerging ones like Mama & Co., to adopt AI-powered digital platforms to remain competitive and meet evolving customer expectations.

AI Chatbots in Food and Beverage E-Commerce

Zipchat AI (2024) noted that AI chatbots integrated into food and beverage e-commerce platforms significantly improve customer engagement by providing instant, automated responses to product inquiries, order tracking, and personalized recommendations. The deployment of AI chatbots reduces the dependency on human customer service staff and ensures that customers receive consistent support at any hour of the day. This finding directly supports the rationale of this study integrating an AI chatbot into the proposed ordering system for Mama & Co. to automate customer communications and reduce manual workload.

Local Literature

Web-Based Ordering System for Start-Up Businesses in the Philippines

Labastida et al. (2024), from the College of Computer Studies at Global Reciprocal Colleges in Caloocan City, Philippines, developed and implemented a web-based ordering system with a forecasting module for a small start-up business that previously relied on social media platforms to sell its products. The system provided business owners with tools to manage orders, track

inventories, and generate sales reports, while also utilizing historical sales data to predict future demand and optimize stock levels. This local study closely mirrors the context of the present research, as both address the limitations of social media-based selling for small businesses and propose a centralized digital ordering platform as a solution.

Development of a Web-Based Food Ordering System

Jiangmin (2024), published in the Asia Pacific Journal of Management and Sustainable Development, developed a web-based food ordering system for a local food establishment that catered to the proprietor, staff, and patrons of the business. The system was designed to provide significant support to restaurant owners in managing their business operations digitally, offering an efficient and accessible platform for both customers and administrators. This study reinforces the relevance and feasibility of developing a web-based ordering system for a local pastry business such as Mama & Co.

Development of a Web-Based Ordering System for a Local Food Business

A study on the development of a web-based ordering system for Angels and Lemons (2024) demonstrated that online ordering methods can effectively alleviate pain points for businesses regardless of their size, simplifying the ordering process by providing necessary product information, enabling order receiving and updating, and including administrative features for efficient management. The system was evaluated using the ISO/IEC 25010 standard and was found to perform well across functionality, reliability, usability, and security dimensions. This local study provides a relevant framework and validation for the development of the proposed ordering system in the present research.

E-Commerce and Digital Transformation of Small Businesses in the Philippines

According to a study on the impact of e-commerce on small and medium enterprises in the Philippines, the COVID-19 pandemic accelerated digital transformation among Filipino businesses, with online shopping users projected to reach 54.7 million by 2024. The study further noted that information and communications technology allows small businesses to collect valuable customer data, streamline business processes such as inventory management and

customer service, and reduce human error through automation. This literature supports the present study's argument that transitioning Mama & Co. from social media-based operations to a structured digital platform is not only timely but necessary for the business to grow and remain competitive in the Philippine digital economy.

Review of Related Systems

This chapter presents a review of related systems, both foreign and local, that are relevant to the development of a Smart web-based ordering system for Mama & Co. The software, platforms, and applications gathered in this chapter serve as the technical benchmark and foundation of the present research.

Foreign Systems

Lulu by SaladStop!

SaladStop! utilizes Lulu, a generative artificial intelligence (GenAI) assistant integrated into their digital ecosystem, allowing customers to personalize and order meals through a chat interface. The system utilizes real-time menu data to ensure high-quality results, allowing customers to input specific dietary exclusions and preferences to instantly generate custom meal options. This system supports the present project's goal of improving customer ordering experience through intelligent, ensuring customers can easily navigate complex menus.

Orderly AI

Orderly is a commercial AI voice-ordering system designed to streamline restaurant order-taking by replacing traditional counter transactions with automated voice agents. This technological platform is designed to handle high-volume rush periods without adding physical hardware, effectively lifting overall sales by freeing up staff. This platform directly supports the rationale of integrating automated inquiry handling to streamline order-taking and reduce manual workload during peak hours, a direct solution to Mama & Co.'s daily order surges.

My Baking Creations (MBC) AI Design Studio

My Baking Creations, a bakery based in the United States, utilizes an online "AI Design Studio" powered by Google Gemini. The parametric builder guides users' layer-by-layer, allowing them to input text descriptions for cake tiers, colors, and toppers, subsequently generating a realistic sketch and a comprehensive PDF design specification sheet for the kitchen staff. This system provides a direct technical framework and validation for the development of the generative AI visualization module in the proposed ordering system, ensuring design accuracy before production.

Pixa AI Cake Generator

Pixa is a dedicated generative AI web tool that creates custom cake designs purely from text prompts. The system allows users to freely specify shapes, number of layers, frosting styles, colors, and specific toppings, producing high-fidelity images of unique cakes ranging from minimalist designs to complex themed creations. This tool underscores the growing need for food businesses, including small and emerging ones like Mama & Co., to adopt AI-powered digital platforms for custom design visualization to remain competitive.

Zakeke Interactive Product Designer

Zakeke is a cloud-based visual commerce platform that integrates with e-commerce websites to offer interactive 2D, 3D, and Augmented Reality (AR) product customization. It enables customers to personalize items and view a realistic representation of their configured product before checkout, automatically generating production-ready files for merchants. This platform supports the rationale of providing customers with real-time visual alignment tools to minimize miscommunication, frequent design changes, and customer dissatisfaction before ordering.

Local Systems

Genie.ph: AI-Powered Custom Cake Marketplace

Genie.ph is the Philippines' first AI-powered marketplace connecting customers with local bakers in Cebu. The platform utilizes an AI engine to analyze user-uploaded reference photos of cakes, breaking down the design into customizable components (icing, toppers) to provide an instant, transparent price quote. This system is highly relevant to the present research as

it demonstrates the practical application of AI in automating custom cake pricing and design, which directly addresses the manual quotation delays and visual misalignment challenges faced by Mama & Co.

ChatGenie: Omnichannel Social Commerce

ChatGenie is a multi-agent AI customer engagement platform that allows Philippine businesses to embed an automated store directly within messaging threads like Facebook Messenger. The platform enables product catalog display, comment auto-reply, auto-confirmation, and complete transactional checkouts without requiring the user to leave the social media application. This platform supports the present project's argument that transitioning Mama & Co. from disorganized manual direct messaging to a structured, automated digital platform is necessary to reduce inquiry latency.

Prosperna E-Commerce Platform

Prosperna is an e-commerce platform specifically built for Philippine SMEs that features built-in AI chatbots alongside standard website building tools. It seamlessly integrates the chatbot directly with the merchant's catalog and inventory, enabling the AI to answer specific product availability questions accurately without human intervention. This system closely mirrors the context of the present research by demonstrating the necessity of providing small local businesses with accessible AI tools to scale their digital sales and automate customer service.

CloudEats: Cloud Kitchen Technology

CloudEats is a pioneer cloud restaurant company operating in the Philippines that utilizes proprietary technology to manage multi-cuisine, multi-brand operations. Their system features deep API integrations with leading food delivery aggregators (GrabFood, FoodPanda), allowing for real-time menu synchronization and centralized kitchen management. This proprietary technology reinforces the relevance of developing a centralized digital platform to efficiently manage multi-brand operations (such as Ben & Mama and 15th Street Kitchen) and delivery integrations for Mama & Co.

Klikit: Restaurant POS and Order Aggregation

Klikit is a cloud POS system engineered specifically for multi-platform restaurants in the Philippines. It natively consolidates orders from GrabFood, FoodPanda, and ShopeeFood into a single unified dashboard, eliminating "tablet fatigue" while providing real-time menu syncing and BIR-compliant digital invoicing. This local POS system closely mirrors the context of the present research, as both address the need for a centralized backend to resolve disorganized order tracking stemming from fragmented delivery platforms.

Ailalay AI-Powered POS

Ailalay provides an integrated, AI-powered operational POS and inventory management system designed specifically for SMEs in the Western Visayas region. The platform features AI workflow automation to reduce manual encoding, alongside robust offline-mode capabilities to ensure continuous operation during internet outages. This software provides a relevant framework for automating inventory tracking and establishing a consistent business workflow that minimizes human error during order processing.

Utak POS

UTAK is a cloud-based tablet POS system tailored for food and beverage businesses in the Philippines. It simplifies operations by offering real-time sales reports, auto-deducting inventory management, a POS-integrated online ordering system, and self-based staff attendance monitoring. This system directly supports the rationale of integrating a reliable backend inventory management module to prevent stockouts and improve the accuracy of stock monitoring practices at Mama & Co.

Synthesis

The reviewed foreign and local related system are collectively validating the technical direction and core feature of the study for Mama & Co. Through all systems reviewed, a recurring theme is the shift toward centralized, automated, and AI enhancing digital platform to address the inefficiencies of manual order management and customer communication

Orderly AI, ChatGenie, and Prosperna are examples of services that can help a business better respond to customer queries and process orders faster. They have great potential for decreasing the manual burden that is typical for the ordering and customer service of Mama & Co. on social media channels. At the same time, the Klikit and CloudEats websites show the importance of using an integrated system that unifies orders coming from various platforms in one dashboard.

As far as the cake design option goes, there are many websites that help to create unique designs. Among them, there are Lulu by SaladStop!, My Baking Creations AI Design Studio, Pixa AI Cake Generator, Zakeke, and Genie.ph. The development of such platforms indicates that consumers want options to view how a particular design will look before making an order. This way, they can better express their needs and avoid confusion or misunderstandings.

In general, all systems analyzed in this review could be helpful to the business in the areas specified. However, they do not seem to provide all functions that should be incorporated into one website for use by a local business operating in a small community. For this reason, a relevant solution to the discussed problem should be proposed.

Current System

Technical Background

The current ordering system by Mama & Co. Customers who order through our social media platforms such as Facebook and Instagram usually begin by sending an inquiry regarding the availability of a product or item. Once the inquiry is received, our Admin Team checks the availability of the requested item based on current stock and production capacity. If the item is available, an order form is sent to the customer to collect the necessary details, including the customer's full name, preferred date and time of pick-up or delivery, complete delivery address, preferred mode of delivery, and mode of payment. After all information has been confirmed and payment has been settled if applicable, the order is officially processed and forwarded to the Kitchen Team for preparation. The kitchen staff carefully prepares and packs the order according to the customer's request and quality standards. Once the order is ready, the customer is notified through direct message. Customers may either pick up the order directly at the store or arrange a third-party delivery service such as GrabExpress or Lalamove for delivery. The transaction is considered complete once the order has been successfully handed over to the customer or delivery rider.

For orders placed through GrabFood and Foodpanda, customers directly place their orders using the respective mobile applications. The orders are automatically received through the GrabFood and Foodpanda merchant portal systems.

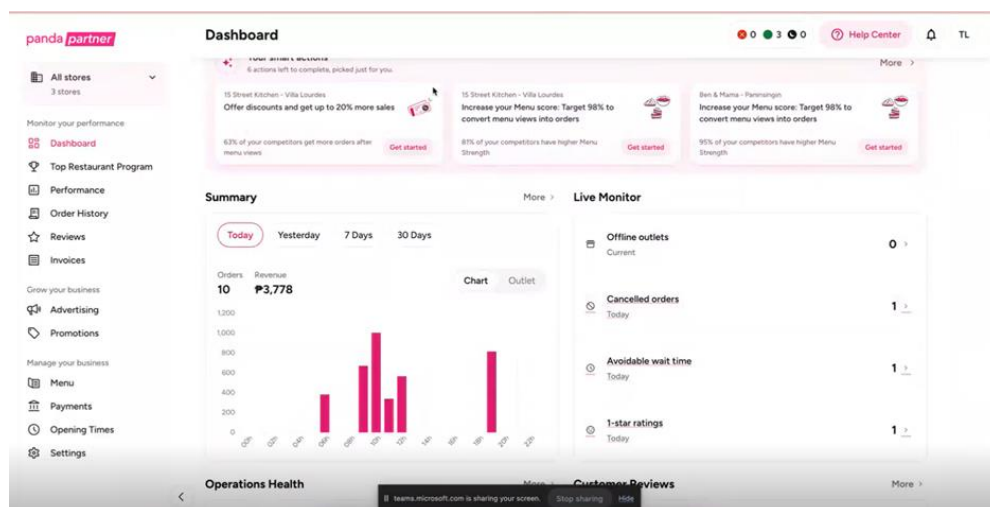


Figure 1: Client's Panda Partner Dashboard

Upon receiving the order, the Kitchen Team first checks the availability of all items before accepting the transaction. If certain items are unavailable, the team may edit the order accordingly or cancel the unavailable items when necessary. Once all items are confirmed available, the order is accepted and prepared based on the quantity and specifications reflected in the system. The kitchen staff ensures that all products are properly prepared, packed securely, and quality-checked before dispatching. After preparation is completed, the staff marks the order as “Ready for Delivery” in the merchant portal, which automatically notifies both the assigned rider and the customer that the order is ready for pick-up. The delivery rider then proceeds to the store to collect the order and deliver it directly to the customer’s doorstep. Throughout the process, customers receive real-time updates through the application until the order is successfully delivered and completed. To maintain strict standardization and ensure delivery quality, management regularly reviews the platform's performance metrics to monitor preparation times, avoidable cancellations, and customer engagement.

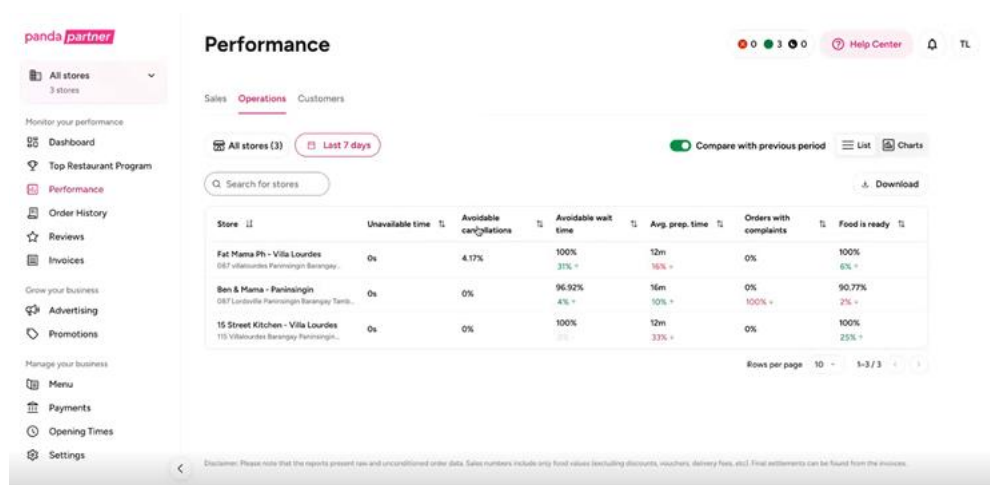


Figure 2: Client's Panda Partner Performance (Operations)

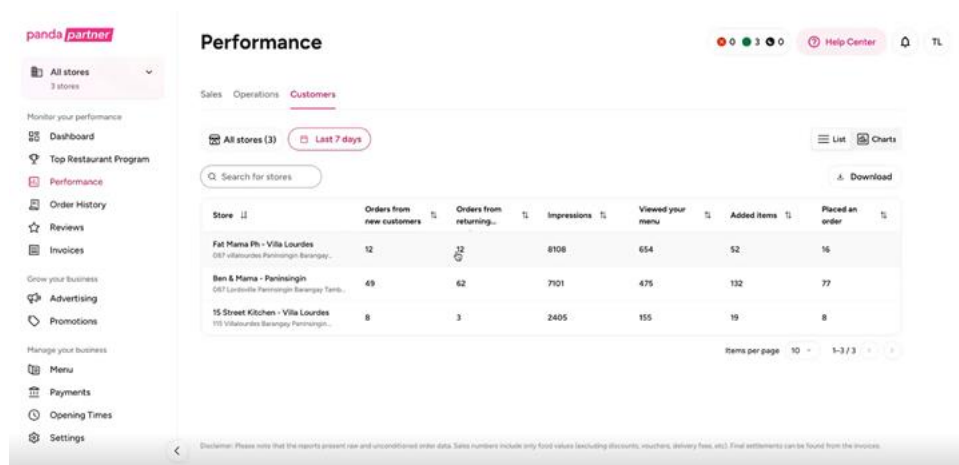


Figure 3: Client's Panda Partner Performance (Customers)

To consolidate orders from both social media and third-party delivery applications, Mama & Co. utilizes DizLog as its central Point-of-Sale (POS) and management system. As multi-channel orders are processed, staff receive real-time notifications in the system to ensure seamless fulfillment.

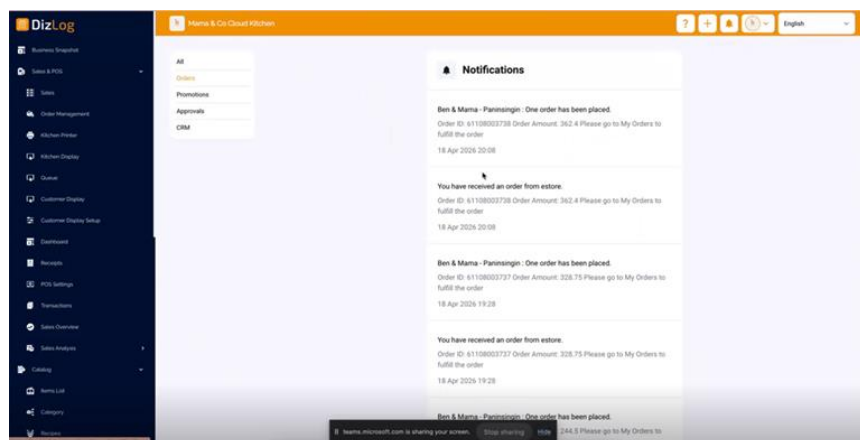


Figure 4: DizLog Dashboard

At the end of the daily operations, management relies on the platform's financial dashboards to track total cash flow, review pending invoices, and generate comprehensive sales summaries across all locations, ensuring total visibility over the business's performance.

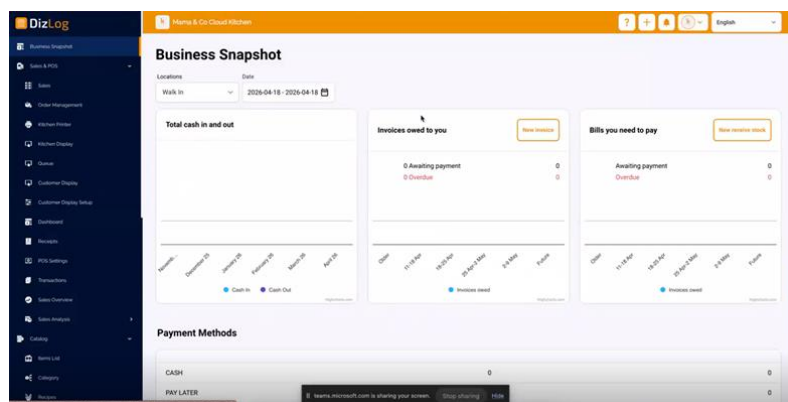


Figure 5: Dizlog Business Snapshot

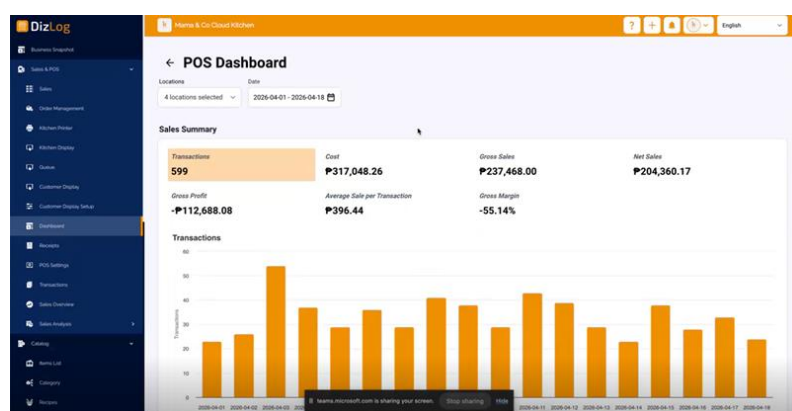


Figure 6: Dizlog POS Dashboard

List of Processes

PROCESS 1: SOCIAL MEDIA ORDERING (FACEBOOK & INSTAGRAM)

Step	What Happens	Who Does It	How Long
1	Customer sends a message on FB or IG asking about product availability or custom order	Customer	Anytime
2	Admin Team gets notification of customer inquiry	Admin Team	Right away
3	Admin checks if items are available based on current stock and production capacity	Admin Team	5-30 min
4	Admin sends order form to customer asking for details (name, delivery date, address, payment method)	Admin Team	5-10 min

5	Customer fills out order form with all required details and confirms	Customer	10-15 min
6	Admin and customer agree on final order details (may make changes to design or specifications)	Both	10-20 min
7	Admin records order details in system (notes/spreadsheet/screenshot) and confirms payment	Admin Team	5-10 min
8	Kitchen Team receives order and starts preparing the cake/product	Kitchen Team	30-120 min (depends on product)
9	Admin notifies customer that order is ready for pick-up or delivery is arranged	Admin Team	2-3 min

Table 1: Social Media Ordering Process

Total Time: From initial message to confirmed order = **1-3 hours**. During busy times (lunch/dinner rush), can take much longer or customer may wait hours just to get a response before they even start making it.

PROCESS 2: GRABFOOD & FOODPANDA ORDERING

Step	What Happens	Who Does It	How Long
1	Customer places order directly through GrabFood or Foodpanda mobile app	Customer	5-10 min
2	Order automatically appears in Mama & Co.'s merchant portal	Platform System	Right away
3	Kitchen Team reviews all items in order and checks if everything is available in inventory	Kitchen Team	2-10 min
4	Kitchen Team accepts order in portal, or edits/removes unavailable items, or declines entire order	Kitchen Team	5-15 min
5	Kitchen Team prepares all items according to order specifications and quantity	Kitchen Team	15-45 min
6	Kitchen Team packs order securely and performs final quality check	Kitchen Team	5-10 min

7	Kitchen Team marks order as "Ready for Delivery" in merchant portal	Kitchen Team	1 min
8	Platform automatically notifies assigned rider that order is ready for pick-up	Platform System	Right away
9	Rider collects order from store and delivers to customer's address; customer receives real-time tracking and notifications	Rider/Platform	20-60 min (depends on distance)
10	Rider marks order as delivered; transaction is recorded in system and platform	Rider/Platform	1 min

Table 2: GrabFood & FoodPanda Ordering

Total Time: From order placement to delivery = **1-2.5 hours** (mostly depends on rider availability and distance).

PROCESS 3: WALK-IN DINE-IN SERVICE

Step	What Happens	Who Does It	How Long
1	Customer enters store and reviews menu	Customer	Anytime
2	Staff member takes customer's order (what product, quantity, any custom requests)	Staff	3-5 min
3	Customer pays for order via cash, card, or GCash	Customer/Cashier	2-3 min
4	Kitchen Team receives order details and starts preparing product	Kitchen Team	15-45 min
5	Staff delivers order to customer (for dine-in) or hands over package (for takeaway)	Staff	2-3 min

Table 3: Walk-In Dine-In Service

Total Time: From order to receiving food = **25-60 minutes** (most is prep time).

SWOT Analysis

Strengths	Weaknesses
• Accessibility	• Manual processes
• Flexibility	• Fragmented orders
• Multi-channel	• Slow response
• Brand presence	• No tracking
• Customer control	• Design mismatch
• Responsiveness potential	• No automation
• Low tech cost	• Poor inventory sync
	• No data
	• Scalability limits
Opportunities	Threats
• Centralized platform	• Competitor tech
• AI design tool	• Rising commissions
• Inventory automation	• Platform risk
• CRM	• Customer expectations
• Analytics	• Staff dependency
• More delivery channels	• Scaling constraints

• Subscriptions	• Data security
• Partnerships	• Bad reviews
• Marketing automation	• Regulations

Table 4: SWOT Analysis

Proposed Solution

Technical Background

The idea behind this system is to put everything in one place. Instead of checking Facebook, then Instagram, then GrabFood separately, everything shows up on a single dashboard. And instead of staff having to answer the same questions over and over (like 'Do you have vegan options?' or 'How much does a 6-inch cake cost?'), a chatbot handles those automatically.

The system has five main parts:

- A chatbot that answers basic questions - When a customer asks something, the bot checks if it's a common question (about prices, ingredients, hours, etc.) and answers right away. If it's something weird or specific, it sends it to a staff member.
- An AI cake design tool - Customers can describe what they want their cake to look like, and the system shows them a picture of what it might look like before they order. This way they don't order something and then be disappointed when they see it.
- One main orders screen - All orders from Facebook, Instagram, GrabFood, and Foodpanda show up in one place, so staff don't have to keep switching between apps.
- Automatic inventory tracking - The system keeps track of what ingredients are in stock and warns staff when something is running low. If an ingredient runs out, the system can suggest similar alternatives.

- Reports and charts - The system keeps track of how many orders come in each day, which cakes are most popular, when the busiest times are, and other useful information for making business decisions.

The system uses common technology that a lot of companies use - basically a website on the backend with a nice interface customers can use on their phone or computer. Everything is stored securely in the cloud, so there's automatic backup in case something goes wrong. The chatbot uses AI to understand what people are asking, and the cake design tool uses AI to generate images.

Feasibility

Operational Feasibility

This solution is operationally feasible as it directly resolves the problem with orders management and customer communication. Owner and business staff manage orders manually on social media (FB and Instagram), where, during working peaks, it becomes impossible to catch up with all messages. With this system implemented, the entire process will become centralized, and repetitive questions will be answered by AI automatically, reducing the manual workloads of the business staff. It will have a simple and creative design to make adoption by both customer and staff easy and fast.

Economic Feasibility

This solution is economically feasible. All development costs of using open source tools and services are quite affordable. Costs for implementing the solution will be mostly related to API charges, however, unlimited tokens will allow generating enough pictures per day to keep expenses reasonable. The long-term savings are connected to reduction of manual labor costs for managing the platform, increasing order precision and growing customer loyalty to the brand.

Technical Feasibility

This solution is technically feasible. All necessary technologies and approaches described above are documented. Integration with the 3rd party APIs will be done through API requests, and deploying this solution on Render will be relatively easy for students' project team. We have the knowledge and programming skills to implement the project.

Schedule Feasibility

The project will be created based on the Agile-Scrum approach, where iterative sprints will be used, matching the academic schedule. Among the major milestones of the system will be requirement analysis and prototyping for the UI/UX part in Phase I, creation of the ordering module and development of a chatbot in Phase II, and implementation of the functionality of cake design by using generative AI in Phase III.

Requirements Analysis

Project Vision

Prototype (Mock Flow / Wireframe)

This section is a Low-Fidelity Wireframe Prototype of the Fatmama.ph from Mama & Co. this shows the list of pages and features that will be implemented to the website.

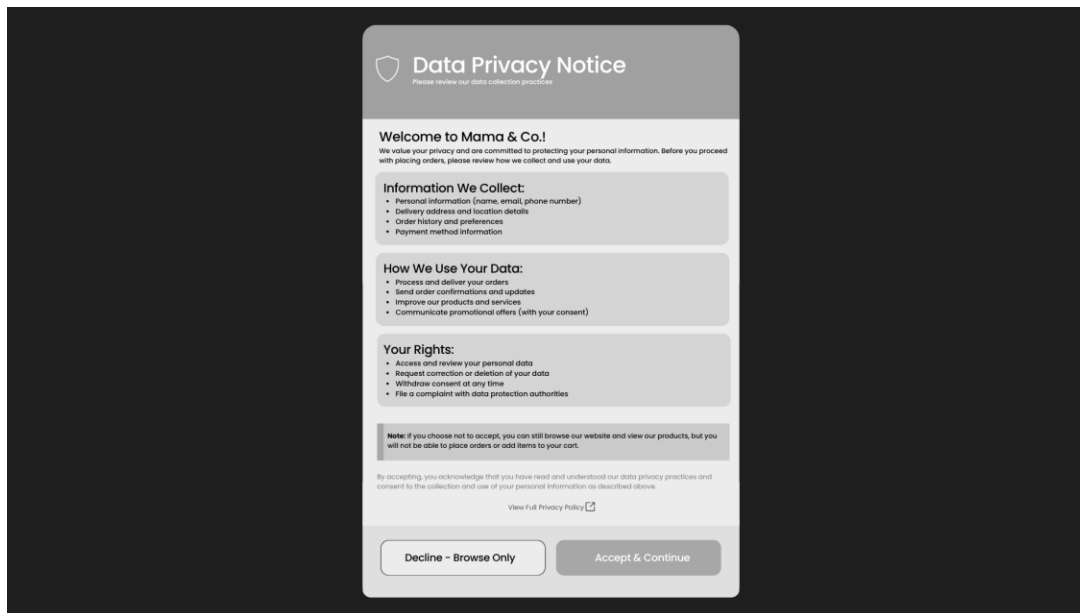


Figure 7: Privacy Notice

Figure 7 demonstrates the data privacy consent when you enter the website.

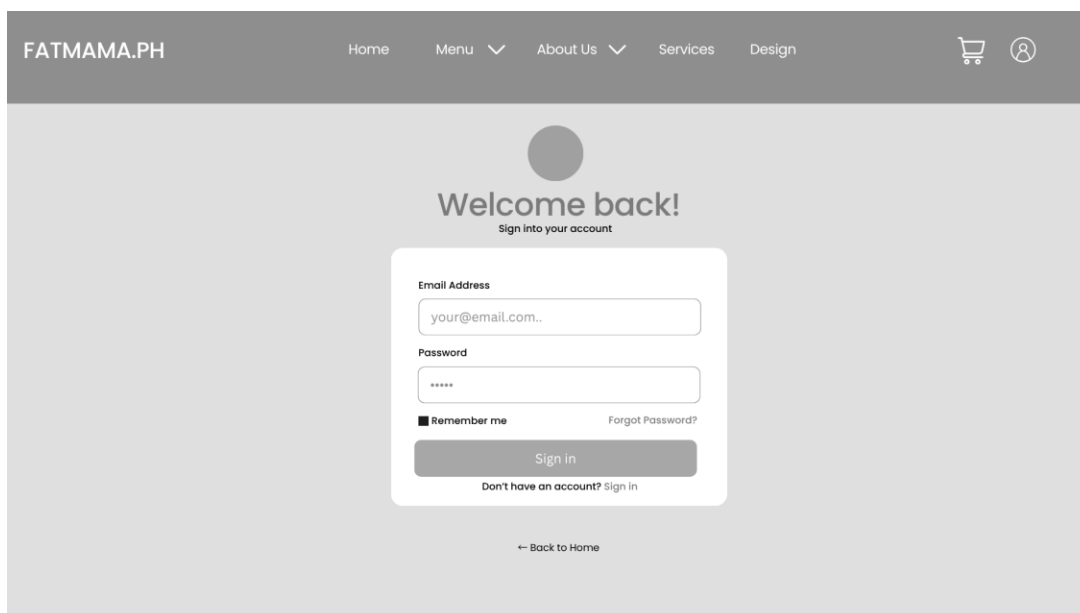


Figure 8: Log in Page

Figure 8 Demonstrates the log in page of the website for users to enter their account.

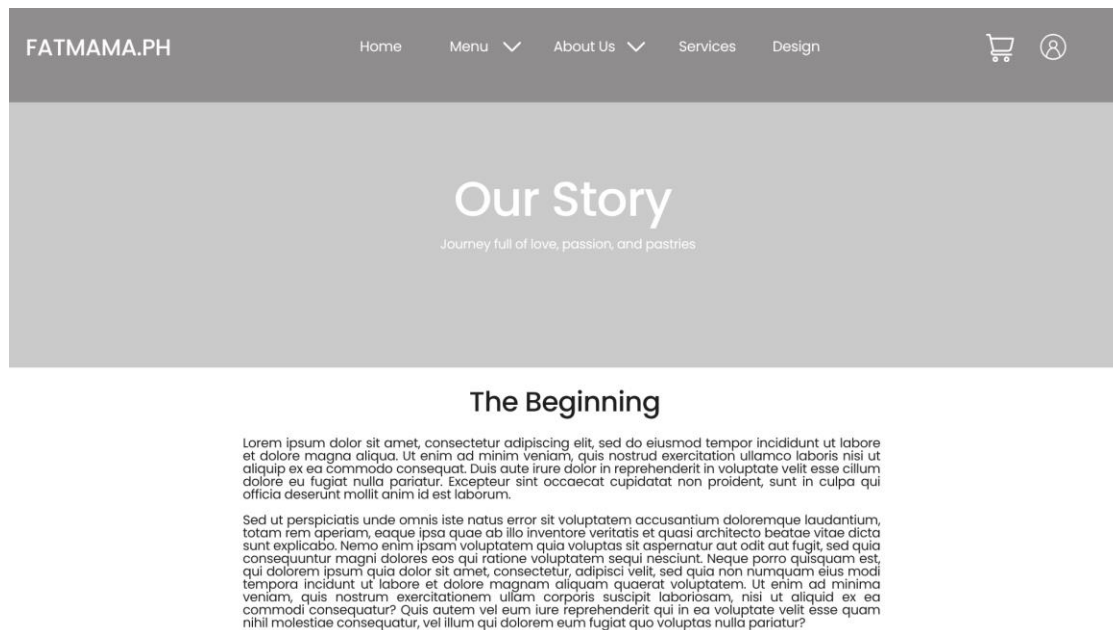


Figure 9: About Us

Figure 9 Demonstrates the story of how Mama & Co. was created, their quality of food etc.

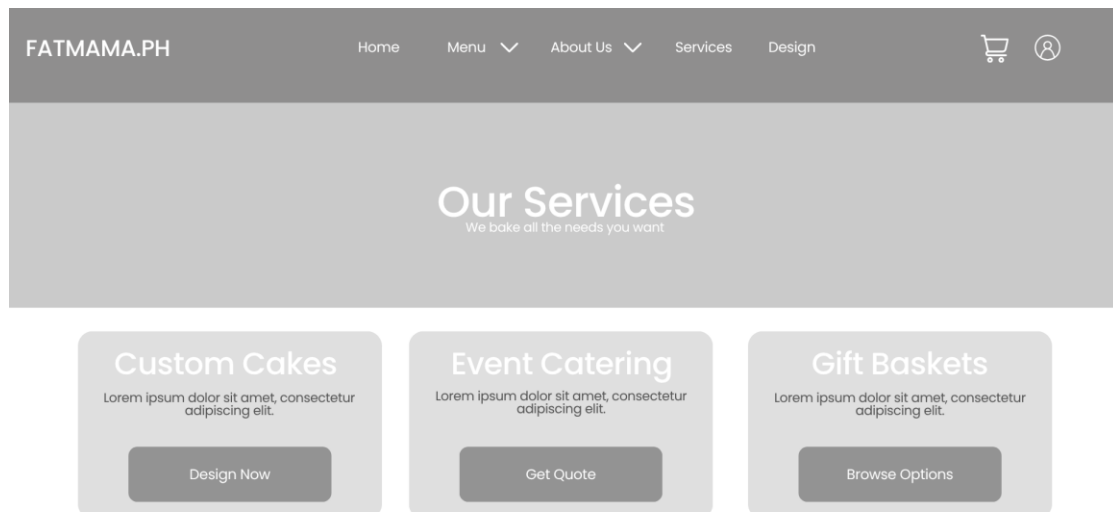


Figure 10: Our Services

Figure 10 Demonstrates the services that Mama & Co. can accommodate.

The screenshot shows the 'AI Cake Designer' section of the website. At the top, the navigation bar includes 'FATMAMA.PH', 'Home', 'Menu', 'About Us', 'Services', and 'Design', along with shopping cart and user icons. The main heading is 'AI Cake Designer' with the tagline 'Create your dream cake with AI-powered customization'. Below this, there are two main panels. The left panel, titled 'Preview', shows a placeholder for a cake image with a large 'X' over it. The right panel, titled 'Customize Your Cake', contains several input fields: 'Cake Text' (with 'Happy Birthday!' entered), 'Size', 'Layers', 'Flavor', 'Frosting', and 'Occasion'.

Figure 11: Cake Designer

Figure 11 Demonstrates the AI Cake designer that Users can use for creating their desired custom cakes.

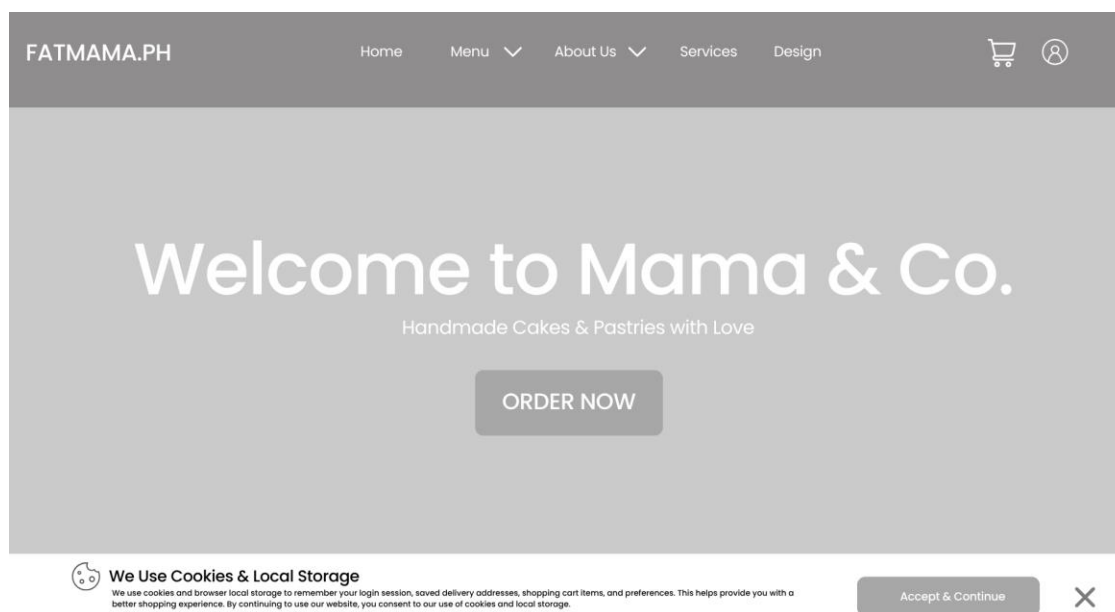


Figure 12: Home Page

Figure 12 Demonstrates the home page of the website.

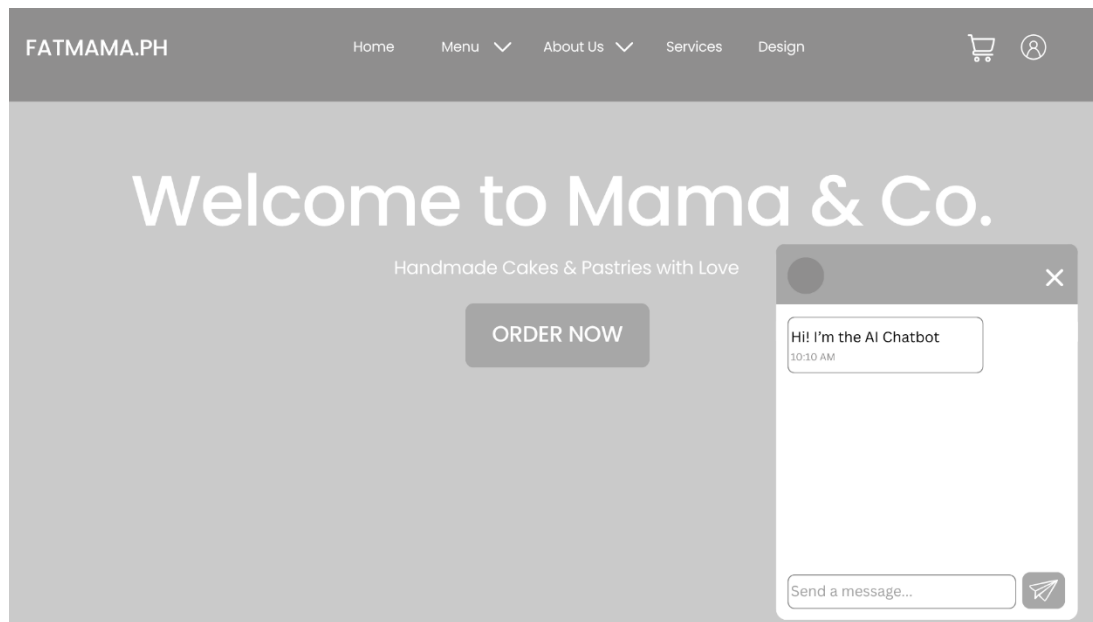


Figure 13: AI Chatbot

Figure 13 Demonstrates the AI Chatbot that users can use to ask FAQs.

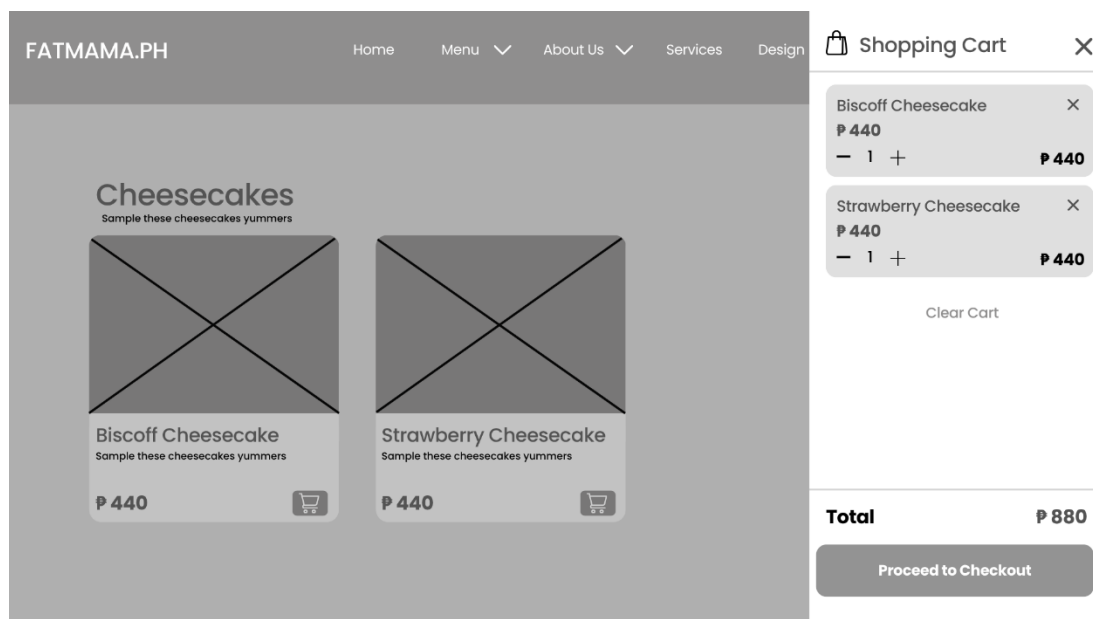


Figure 14: Shopping Cart

Figure 14 Demonstrates the shopping cart of the website used for placing an order.

FATMAMA.PH
Home
Menu
About Us
Services
Design

Checkout

Delivery Information

First Name:

Last Name:

Email:

Phone No.:

Address

City

Province

Zip Code

Delivery Notes (Optional)

Order Summary

Biscoff Cheesecake	₱ 440
Qty: 1	
Strawberry Cheesecake	₱ 440
Qty: 1	
Subtotal	₱ 880
Delivery Fee	₱ 30
Total	₱ 910

Place Order

By placing this order, you accept to our Terms and Conditions

Figure 15: Checkout Page

Figure 15 Demonstrates the checkout page for users to provide customer details etc.

FATMAMA.PH
Home
Menu
About Us
Services
Design

Delivery Notes (Optional)

☒ Save this delivery address for future orders

Payment Method

☒ **Cash on Delivery**
Pay when you receive the order

☐ **GCash**
Pay via GCash e-wallet

☐ **PayMaya**
Pay via PayMaya e-wallet

Strawberry Cheesecake ₱ 440
Qty: 1

Subtotal ₱ 880
Delivery Fee ₱ 30

Total ₱ 910

Place Order

By placing this order, you accept to our Terms and Conditions

Figure 16: Checkout Page

Figure 16 Demonstrates the checkout page for the delivery notes and payment method.

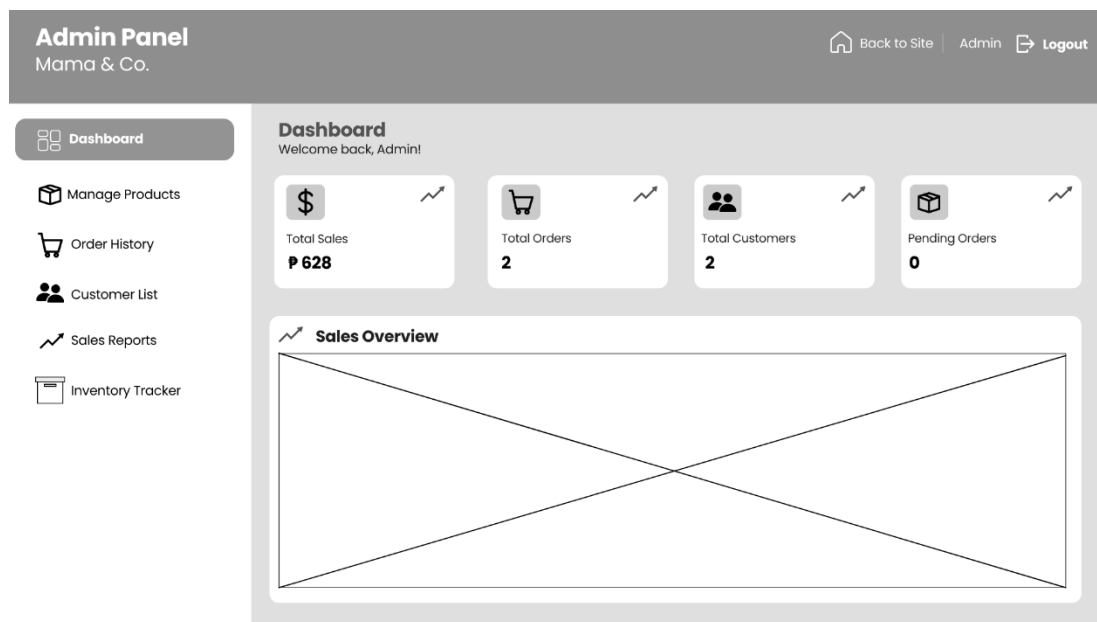


Figure 17: Admin Panel Dashboard

Figure 17 Demonstrates the admin panel dashboard where it summarizes the sales reports, customer list etc.

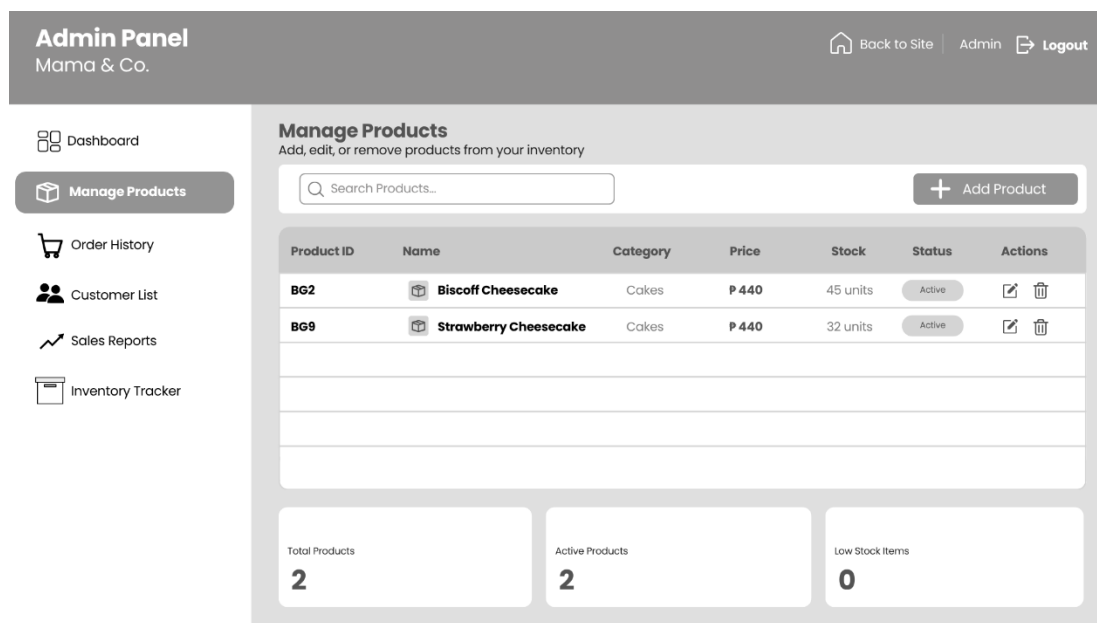


Figure 18: Manage Products Page

Figure 18 Demonstrates the manage products page where the admin can manage products to sell.

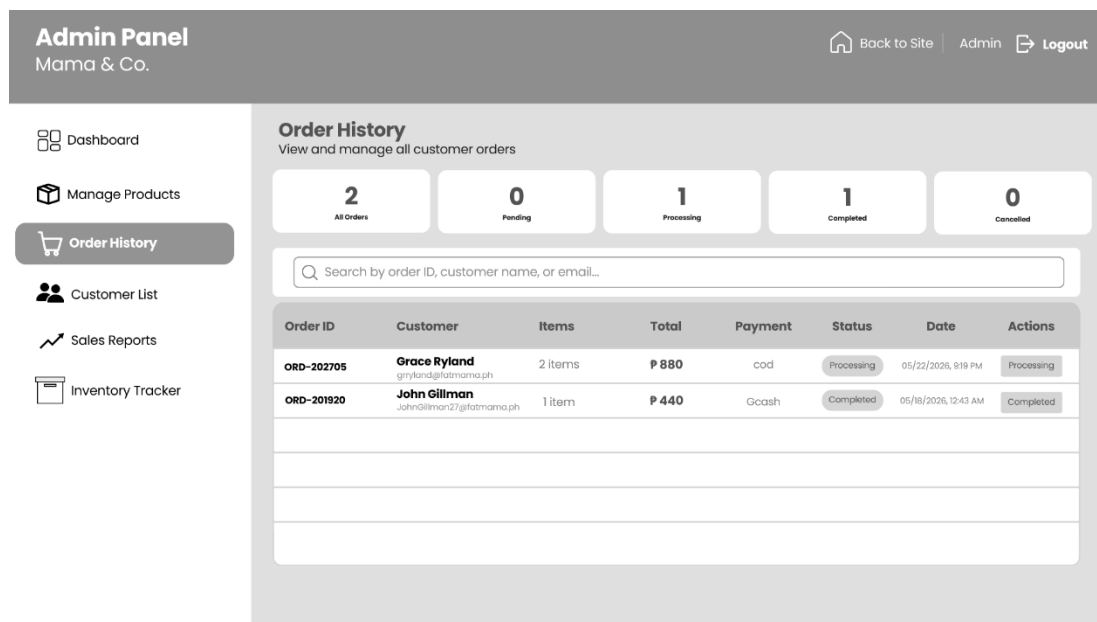


Figure 19: Order History Page

Figure 19 Demonstrates the order history page where the admin can see all the customers orders.

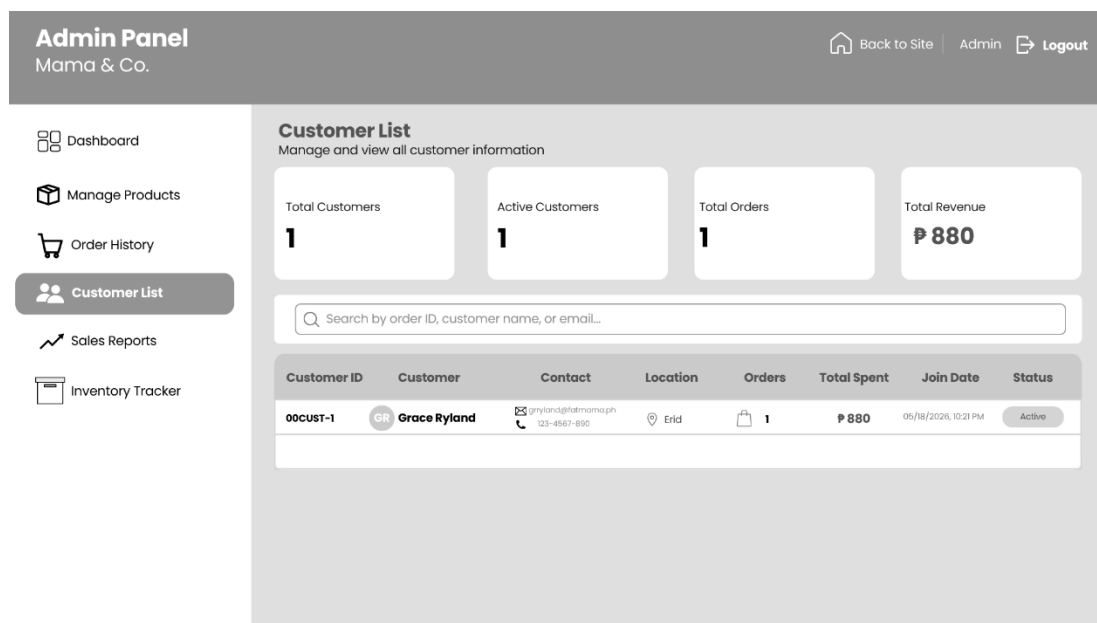


Figure 20: Customer List Page

Figure 20 Demonstrates all the customer information etc.

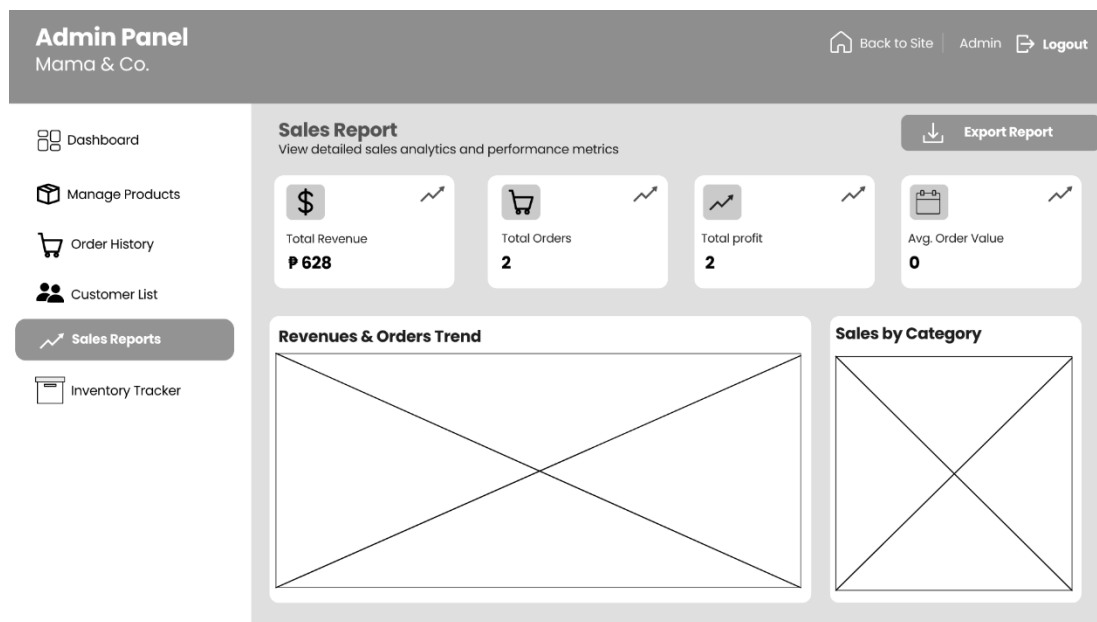


Figure 21: Sales Reports Page

Figure 21 Demonstrates the sales reports page where the admin can see all the reports/

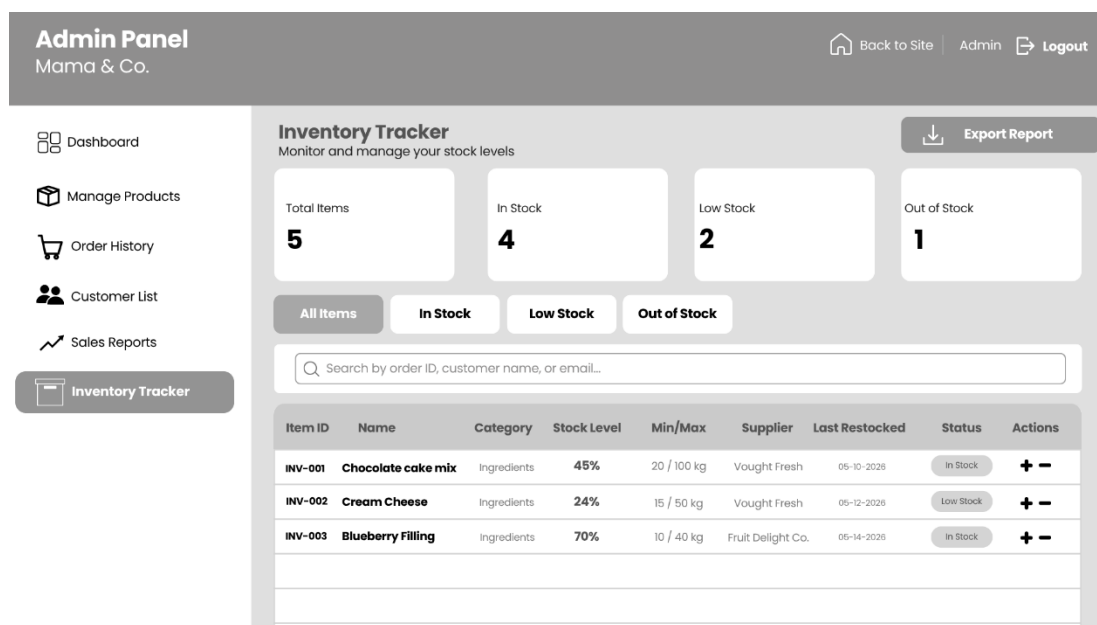


Figure 22: Inventory Tracker Page

Figure 22 Demonstrates the inventory tracker page where the admin see the stock level of materials and ingredients.

Project Lean Canvas

Problem

1. **Difficulty in Order Tracking:** Scattered chat messages make it hard to record, monitor, and update orders, leading to inconsistent records and difficulty in managing the daily volume of customers
2. **Time-Consuming Inquiry Handling:** The process of responding to customer inquiries is time-consuming due to repetitive questions that require manual responses every time.
3. **Customization mismatch:** Not being able to preview the cake design often causes confusion, frequent changes, and customer dissatisfaction with their orders.
4. **Problem in Inventory Tracking:** Without proper inventory tracking, the pastry shop's ordering system becomes prone to order errors and inconsistencies in product availability and customer service.
5. **Difficulty in Report Generation:** The manual system makes report generation time-consuming and error-prone, resulting in delays and unreliable information.

Solution

1. **AI Chatbot:** Uses NLP to automate 24/7 responses to common queries and filter ingredient exclusions.
2. **Generative AI Design:** Powered by DALL-E to provide instant visual previews of custom cake orders.
3. **Centralized POS Portal:** Consolidates orders from social media and third-party apps like Foodpanda/GrabFood.

Existing Alternatives

1. Placing orders manually by sending direct messages through Facebook and Instagram.
2. Standard local competitors (e.g., Contis, Mary Grace) lack instant customization previews.

Key Metrics

1. Response time reduction (Target: >50%).

2. Order accuracy/reduction in design-related changes.
3. Daily order volume and retention metrics.

Unique Value Proposition

1. A professional web system that automates Lipa City's most popular cloud kitchen, replacing slow social media chats with 24/7 AI support and real-time cake previews.
2. One seamless checkout for all your cravings. Mix and match premium cakes, savory dishes, and everyday bakes from Fatmama.ph in a single order.
3. What you see is exactly what you get. Our Generative AI preview bridges the gap between your ideas and our bakers, ensuring 100% design accuracy and zero party-day disappointments.

High-Level Concept

A 24/7 Virtual Pastry Assistant that designs, quotes, and takes orders while the staff focuses on baking.

Unfair Advantage

1. Direct Integration of multi-brand management (Mama & Co., Ben & Mama, 15th Street Kitchen) with localized AI-driven inventory and POS tracking specific to the Lipa City market.
2. Proprietary Cake Design & Inventory Integration
3. Proprietary Pastry NLP for Local Tastes

Channels

1. The business's existing Facebook and Instagram social media pages.
2. recommendations between friends and regular customers.
3. Physical dine-in store in Villa Lourdes.

Customer Segments

1. Online food buyers
2. Ages 22 - 28
3. Middle - income Family

Early Adopters

1. Customers on budget.
2. Existing Customers
3. Customers looking for highly customized cakes for specific events.

User Classes and Characteristics

There are basically two types of people who'll use this system: customers ordering cakes, and the staff at Mama & Co. managing everything.

Online Customers

	Details
Who they are	Mostly young people, families, and couples in and around Lipa City
What they want	Easy ordering, see what the cake looks like before they buy it
Tech skills	Pretty comfortable with phones and websites
What they need	Quick ordering, design preview, track when their order arrives, pay easily
Time on the site	Usually 5-15 minutes per visit
Phone or computer	Mostly phones (about 2/3 use phones, 1/3 use computers)
Frustrations with current system	Can't see what custom cakes look like, slow response times on messenger, confused about pricing
What this gives them	Faster ordering, they see designs before buying, fewer annoying back-and-forths

Table 5: Online Customers

Basically, here's what customers will do when they use the system:

- Browse what Mama & Co. has available
- Ask the chatbot questions if they're confused about something
- Pick items and customize them (pick flavors, size, any special diet stuff)
- Use the AI to describe their cake idea and see what it looks like
- If they want to change the design, they can describe it again
- Pay for the order
- Get a confirmation and know roughly when it'll be ready
- Check the status as they wait
- Get points for future discounts on their next order

Shop Owner and Staff

	Details
Job titles	Owner, couple managers, kitchen staff, delivery people

What they want	Organized orders, know where inventory stands, see what's selling
Tech skills	Owner and managers are okay with tech, kitchen staff less so
What they need	See all orders in one place, track inventory, create sales reports, communicate with team
Daily time on system	Owner checks a few hours daily, managers most of the day, kitchen staff when needed
Phone or computer	Mostly computers and tablets at the shop
Frustrations with current system	Orders scattered everywhere, inventory goes missing, takes forever to make reports
What this gives them	Everything in one view, less manual data entry, quick reports they can actually read

Table 6: Shop Owner and Staff

On the staff side, they'll be doing things like:

- Log in and see all orders that came in from everywhere
- Check if there's enough ingredients to make each order
- Accept or tell the customer an order isn't possible
- Look at cake design pictures the customer or AI made and either approve them or ask for changes
- Handle any weird questions that the chatbot can't figure out
- Mark orders as 'making it', 'done', and 'delivered'
- Keep track of ingredients and get a heads-up when something's getting low
- At the end of the day or week, see reports about how many cakes sold, which ones are popular, etc.
- Coordinate with the kitchen about what to make next

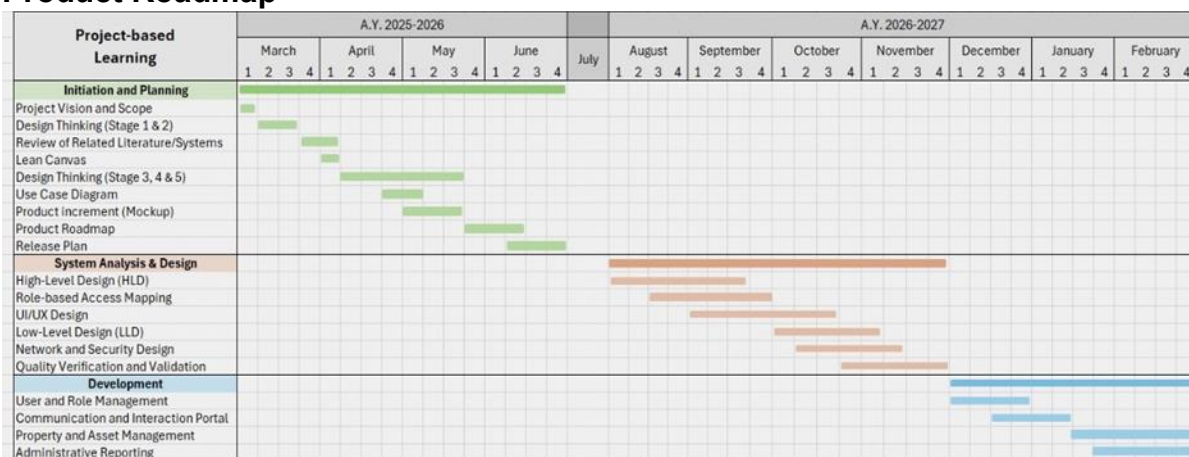
E. Product Backlog / User Stories

ID	As a ...	I want to be able to ...	So that ...
1	Business Owner	view all incoming orders from multiple channels in one unified dashboard	I can have full visibility of all customer orders regardless of the sales channel

2	Staff Member	accept or reject incoming orders with a single click	I can quickly manage order flow and respond to customers in real-time
3	Kitchen Staff	see the current status of orders and update them (in-prep, ready, completed)	staff can coordinate efficiently and ensure no orders are forgotten
4	Business Owner	view real-time order statistics and analytics	I can make informed decisions about peak hours, staffing, and inventory
5	Delivery Driver	view assigned orders and mark them as picked-up and delivered	the system can track deliveries in real-time and customers receive updates
6	Staff Member	receive automated alerts when orders exceed normal volume	we can prepare for high-demand periods and ensure service quality
7	Customer	receive real-time order status notifications	I always know when my order is being prepared and when it will be ready
8	Cake Designer	view cake design customization requests and templates in the system	I can efficiently manage custom design orders and provide quality work

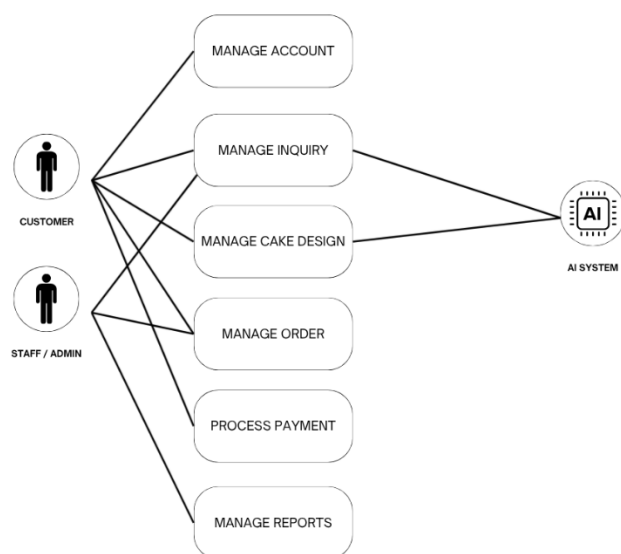
Table 7: Product Backlog / User Stories

Product Roadmap



Use Case Diagram

SMART WEB ORDERING SYSTEM WITH
CHATBOT ASSISTANCE AND
GENERATIVE CAKE DESIGN
USE CASE DIAGRAM
(HIGH LEVEL)



The use case diagram for the Smart Web Ordering System with Chatbot Assistance and Generative Cake Design. There are three main actors in the system: the customer, staff/admin, and the AI system. Customers use the platform to inquire about products via the AI chatbot, visualize custom cakes using the generative AI tool, and place orders directly on the website. Staff members manage the backend of the ordering system. They use the centralized POS portal to confirm item availability, handle complex inquiries routed from the chatbot, and manage orders consolidated from various platforms like Facebook, FoodPanda, and GrabFood.

Additionally, the owner uses the system to monitor overall business performance through a statistics and analytics dashboard. Finally, the AI system acts as an AI chatbot with NLP capabilities that automatically answers common customer queries, as well as a generative AI cake design feature that produces visual previews of custom orders based on customer-provided preferences.

Use Full Description

Conclusion

Right now, Mama & Co. is held back by how scattered everything is. Orders come through different apps, inventory is tracked by hand, and staff spend a lot of time answering the same questions over and over. This system fixes those problems by putting everything together, automating the repetitive stuff, and giving customers a way to actually see what their cake will look like.

It makes sense to build this from all angles. Staff can definitely use it and learn it. It costs money upfront, but you get way more back in the first year. The technology exists and our team knows how to use it. And we can build it in a reasonable timeline.

When it's all set up, good things should happen pretty quickly: customers get their orders confirmed faster (about 50% faster), fewer mistakes happen because customers can see the design upfront, the team has way less manual data entry work, and more people might order because the experience is better.

The whole project takes about 4.5 months from start to finish, with the main features ready by month 3. Then we stick around for a month after launch to make sure everything's working and the team's comfortable.

This is the kind of thing that could work for other small cake and pastry shops in the Philippines too. But first, let's get Mama & Co. running smooth with it.

Right now, Mama & Co. is held back by how scattered everything is. Orders come through different apps, inventory is tracked by hand, and staff spend a lot of time answering the same questions over and over. This system fixes those problems by putting everything together, automating the repetitive stuff, and giving customers a way to actually see what their cake will look like.

It makes sense to build this from all angles. Staff can definitely use it and learn it. It costs money upfront, but you get way more back in the first year. The technology exists and our team knows how to use it. And we can build it in a reasonable timeline.

When it's all set up, good things should happen pretty quickly: customers get their orders confirmed faster (about 50% faster), fewer mistakes happen because customers can see the design upfront, the team has way less manual data entry work, and more people might order because the experience is better.

The whole project takes about 4.5 months from start to finish, with the main features ready by month 3. Then we stick around for a month after launch to make sure everything's working and the team's comfortable.

This is the kind of thing that could work for other small cake and pastry shops in the Philippines too. But first, let's get Mama & Co. running smooth with it.

References

- [1] R, J. W. (2025). The role of social media and online platforms in growing home baking business. *International Journal for Research in Applied Science and Engineering Technology*, 13(4), 3065–3073. <https://doi.org/10.22214/ijraset.2025.68509>
- [2] Prince, L., Marvile, S., John, E. C., Wiljohn, F., & Centeno, C. (2024). Web-based ordering system for Start-UP business with forecasting. *World Journal of Advanced Research and Reviews*, 22(3), 357–368. <https://doi.org/10.30574/wjarr.2024.22.3.1721>
- [3] Cloudwaitress. (2025, September 14). *What is an online ordering system?* <https://www.cloudwaitress.com/online-ordering/what-is-an-online-ordering-system/>
- [4] Merriam-Webster. (2026, April 9). Chatbot. In *Merriam-Webster.com dictionary*. <https://www.merriam-webster.com/dictionary/chatbot>
- [5] Merriam-Webster. (2026, April 3). Artificial intelligence. In *Merriam-Webster.com dictionary*. <https://www.merriam-webster.com/dictionary/artificial%20intelligence>
- [6] Merriam-Webster. (2026, April 8). Generative AI. In *Merriam-Webster.com dictionary*. <https://www.merriam-webster.com/dictionary/generative%20AI>
- [7] Merriam-Webster. (2026, April 5). System. In *Merriam-Webster.com dictionary*. <https://www.merriam-webster.com/dictionary/system>
- [8] Merriam-Webster. (2026, February 24). Natural language processing. In *Merriam-Webster.com dictionary*. <https://www.merriam-webster.com/dictionary/natural%20language%20processing>
- [9] Uzoka, A., Cadet, E., & Ojukwu, P. U. (2024). Leveraging AI-powered chatbots to enhance customer service efficiency and future opportunities in automated support. *Computer Science & IT Research Journal*, 5(10), 2485–2510. <https://doi.org/10.51594/csitrj.v5i10.1676>
- [10] Yasar, K. (2024, November 25). *What is web application (web apps) and its benefits?* TechTarget SearchSoftwareQuality. <https://www.techtarget.com/searchsoftwarequality/definition/Web-application-Web-app>
- [11] Oracle. (n.d.). *What is cloud kitchen?* <https://www.oracle.com/ae/food-beverage/cloud-kitchens/>
- [12] National Privacy Commission. (2012). *Data Privacy Act of 2012* (Republic Act No. 10173). <https://privacy.gov.ph/data-privacy-act/>

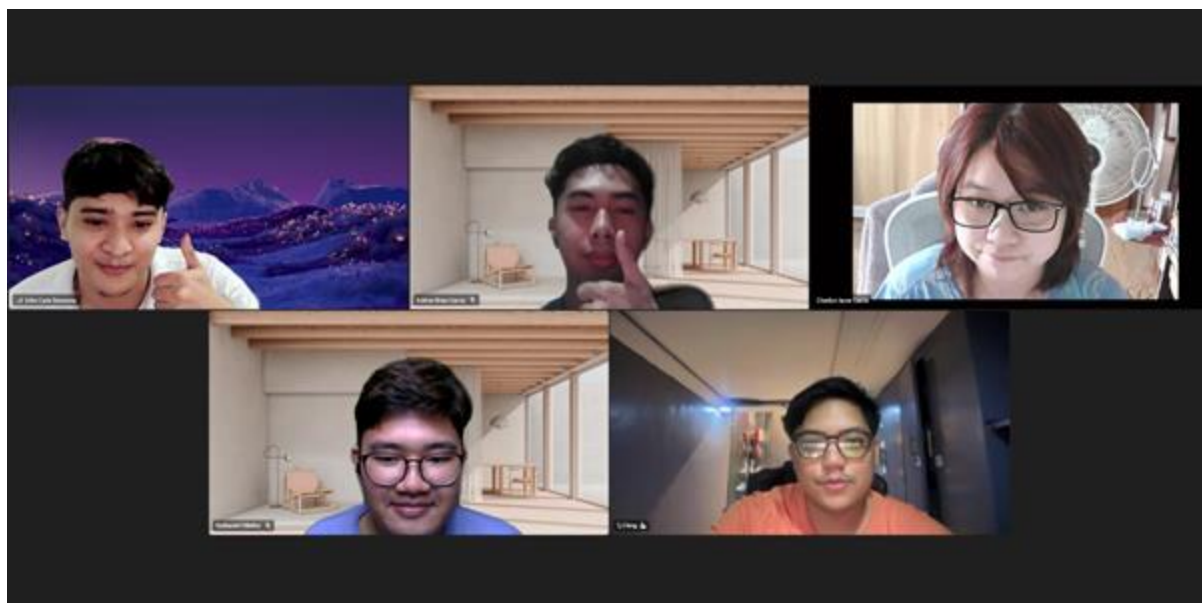
- [13] Jiangmin, L. (2024). Development of a web-based food ordering system. *Asia Pacific Journal of Management and Sustainable Development*, 12(2), 155–159. <https://asiapijournals.org/wp-content/uploads/2024/11/15.-APJMSD-2024-67.pdf>
- [14] De Vera, J., et al. (2024). Development of a web-based ordering system for Angels and Lemons. *International Journal of Advanced Multidisciplinary Research*, 8(6), 77–83. <http://ijeais.org/wp-content/uploads/2024/6/IJAMR240606.pdf>
- [15] Mendoza, J., et al. (2023). The impact of e-commerce on small and medium enterprises in the second district of Albay. *International Journal of Research and Innovation in Social Science*. <https://rsisinternational.org/journals/ijriss>
- [16] Gupta, M., Dheekonda, V., & Masum, M. (2024). Genie: Enhancing information management in the restaurant industry through AI-powered chatbot. *International Journal of Information Management Data Insights*, 4(2), Article 100255. <https://doi.org/10.1016/j.jjime.2024.100255>
- [17] Yaiprasert, C., & Hidayanto, A. N. (2024). AI-powered in the digital age: Ensemble innovation personalizes the food recommendations. *Journal of Open Innovation: Technology, Market, and Complexity*, 10(2), Article 100261. <https://doi.org/10.1016/j.joitmc.2024.100261>
- [18] Jenkins, A. (2024, October 11). *15 ways AI is impacting the restaurant industry*. Oracle NetSuite. <https://www.netsuite.com/portal/resource/articles/business-strategy/ai-in-restaurants.shtml>
- [19] Zipchat AI. (2024). *AI chatbot for food and beverage e-commerce store*. <https://www.zipchat.ai/post/ai-chatbot-on-your-food-and-beverage-ecommerce>
- [20] D. Magbanua, “What to know about SaladStop!’s new AI assistant, Lulu,” *Asian Business Review*, Nov. 04, 2025. [Online]. Available: <https://asianbusinessreview.com/food-beverage/exclusive/what-know-about-saladstops-new-ai-assistant-lulu>

Appendices

Appendix A: Roles and Responsibilities

Name	Roles
Charlize Jacee Garcia	• UI/UX Designer • Mock-up/Wireframe • Statement of the Problem • Documentation
Ashton Brian Garcia	• Documentation • Video teaser • Client Call • RRS
John Carlo Baracena	• Documentation • PPT • Researching • Debugging • RRL
Nathaniel Villaflor	• Documentation • Product Teaser • Prototype • RRL

Appendix B: Minutes of the Meetings



First meeting with the client



First meeting with the Project Adviser

Appendix D: Project SharePoint Link

<https://asiapacificcollege.sharepoint.com/:f:/s/FATMAMA.PH/IgA0NZiJV6sPT4AxeTiMhs0FAczwm5Fvsdc403zEZibL72I?e=ypGnMH>

Appendix E: Requirements Traceability Matrix

Req ID	User Story / Feature Description	Source / Role	Technical Component / Module	Verification Method
REQ-01	View all incoming orders from multiple channels in one unified dashboard.	Business Owner	Centralized POS Portal & Dashboard Component (React / Node.js)	Integration testing with mock platform APIs
REQ-02	Accept or reject incoming orders with a single click.	Staff Member	Order Management Backend Controller (Node.js / PostgreSQL)	Unit testing of order status transition states
REQ-03	Track and update order production status (In Prep, Ready, Completed).	Kitchen Staff	Production Workflow Component & Real-time Database Sync	UI walkthrough and database status check
REQ-04	AI Chatbot with NLP to automatically answer 24/7 common queries and handle ingredient exclusions.	Online Customer / Staff	NLP Chatbot Engine (OpenAI API integration)	Intent classification accuracy and response latency checks
REQ-05	Generative AI cake design feature to	Online Customer	DALL-E Generative AI	Prompt testing inside the

	provide instant visual previews from text prompts.		Engine Integration	boundary constraints of Mama & Co.
REQ-06	Automatic inventory tracking and out-of-stock alternative ingredient suggestions.	Shop Owner / Staff	AI-Driven Inventory Management Module	Boundary testing for low stock alerts and algorithm trigger
REQ-07	Statistics and analytics dashboard tracking peak seasons, order volume, and retention.	Business Owner	Data Visualization / Analytics Dashboard (React Charts)	SQL aggregate query performance and UI rendering test
REQ-08	Implement data privacy controls (consent checkboxes, password hashing) compliant with RA 10173.	System / Legal Compliance	Authentication & Security System (Bcrypt, Cookie Banner)	Security code audit and manual verification of checkbox states